

HARVESTHUB – FARM MANAGEMENT SYSTEM

Software Requirements Specification (IEEE Std 830-1998 Compliant)

Comprehensive Crop Rotation, Field Operations, Financial & Maintenance Tracking Platform

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Program: HCL Industrial Training Program (Session 2026–27)
Project ID: P_008 | **Domain:** Agricultural Engineering / Full Stack

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document provides a comprehensive, rigorous definition of the functional, non-functional, data, and architectural requirements for **HarvestHub**. Designed as a web-based enterprise solution for agricultural management, HarvestHub digitizes field-level planning, algorithmic crop rotation, operational assignment, cost and yield analysis, and preventive equipment maintenance. This SRS baseline serves as an official technical agreement for development, academic evaluation, and quality assurance under IEEE 830-1998 standards.

1.2 Document Revision & Approval History

Version	Date	Primary Changes	Prepared By	Approved By
0.1	15-Sep-2026	Initial draft, scope boundary definition, and requirements capture.	Saksham Srivastava	HCL Project Guide
1.0	30-Sep-2026	Final baseline SRS complete with UML, DFD, ERD, and API contracts.	Saksham Srivastava	HOD / Supervisory Panel

1.3 Product Scope & Key Objectives

HarvestHub bridges the gap between traditional farm management and data-driven decision making. The primary domain objectives include:

- Field-Level Digital Mapping:** Subdivide multi-acre agricultural properties into logical fields annotated with soil types (e.g., Alluvial, Black, Clay, Loamy) and area metrics.
- Algorithmic Crop Rotation:** A rule-based engine that evaluates historical crop sequences to prevent mono-cropping, minimize soil nitrogen depletion, and maximize yield.
- Operations & Resource Scheduling:** Automated task creation for land preparation, sowing, irrigation, fertilizing, spraying, and harvesting with worker assignments.
- Financial & Profitability Analytics:** Granular tracking of input expenditures vs. yield harvest value, computing net revenue, cost per acre, and yield per acre.
- Preventive Maintenance Engine:** Equipment service tracking with automated daily cron jobs flagging machines as DUE or OVERDUE with email alerts.

System Scope Boundaries

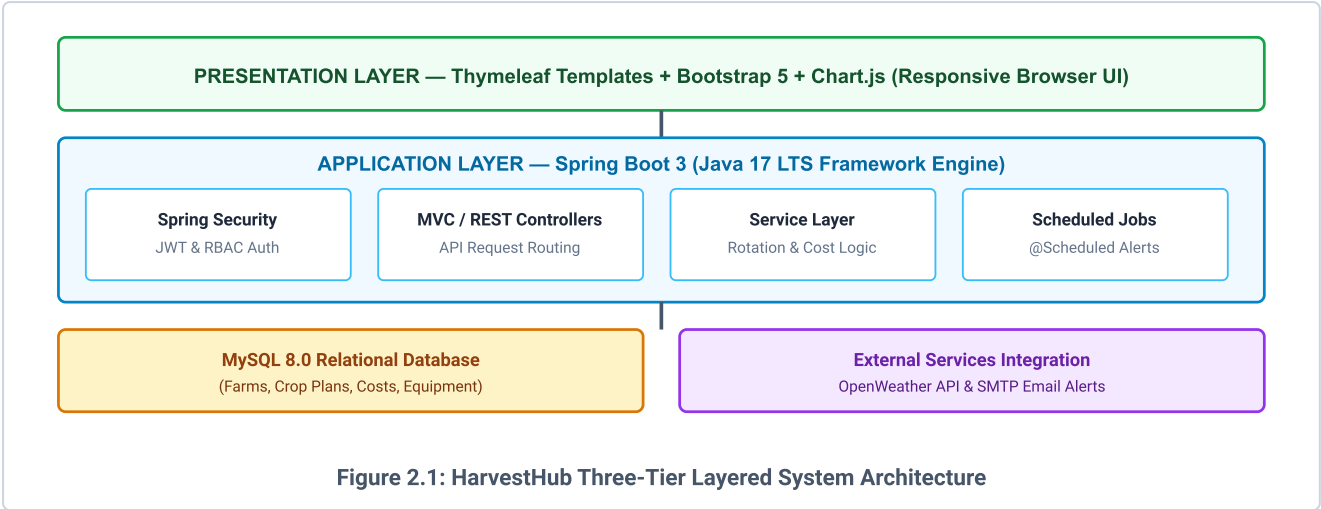
In-Scope: Multi-farm mapping, rule-based crop recommendation, operation calendar, financial ledger, equipment maintenance, REST APIs, and role-based access.

Out-of-Scope (Future Release): Direct IoT hardware sensor mesh integration, live satellite imagery processing, and real-time payment gateway settlement.

2. System Architecture & Overall Description

2.1 High-Level System Architecture

HarvestHub is structured around a three-tier layered architecture enforcing modular separation of concerns. The presentation layer utilizes Thymeleaf with Bootstrap 5 and Chart.js. The backend application layer is powered by Java 17 LTS and Spring Boot 3, leveraging Spring Security with JWT tokens for stateless authentication. Persistence is governed by Spring Data JPA (Hibernate) connecting to a MySQL 8.0 relational database.



2.2 User Classes & Responsibilities

User Role	Access Privilege	Core Operational Responsibilities
Farm Owner	Full Read / Write	Registers properties, defines fields, monitors high-level profit/loss reports, manages farm managers and billing profiles.
Farm Manager	Operational Control	Configures crop plans, executes crop rotation suggestions, assigns field operations, logs costs and harvests, manages equipment.
Field Worker	Task Restricted	Accesses assigned field activities, marks tasks as In-Progress or Completed, logs field notes.
System Admin	Administrative	Maintains master seed catalogues, crop rotation rules, system logging, user authorization overrides.

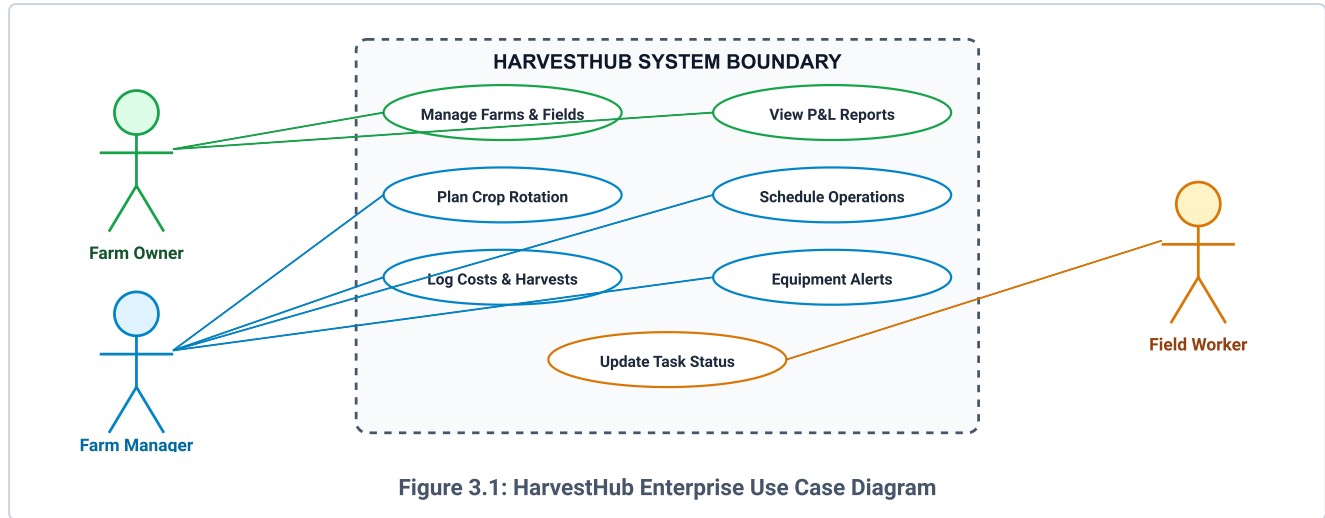
2.3 Technical Environment & Operating Constraints

- Runtime Environment:** Java Development Kit (JDK 17.0.x LTS) with Spring Boot 3.2.x framework.
- Database Server:** MySQL Server 8.0.x configured with InnoDB engine and ACID transaction compliance.
- Security Constraints:** BCrypt password hashing (\$2a\$10\$), JWT stateless session tokens (120-minute expiry), HTTPS/TLS 1.3 encryption.
- Hosting & Containerization:** Docker single-container / Docker-Compose orchestrating App and MySQL instances.

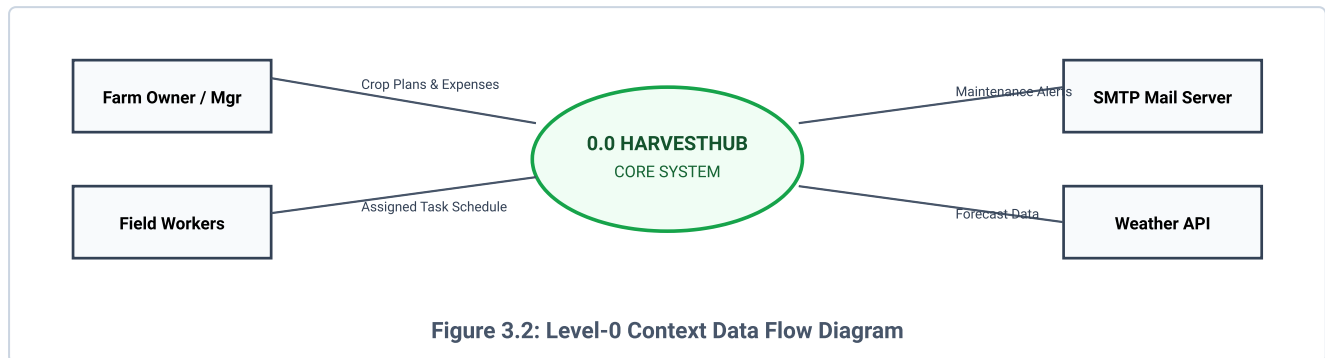
3. System Behavioral Models & Functional Requirements

3.1 Use Case Diagram

The Use Case diagram below maps the primary system actors to their core operational interactions within the HarvestHub system boundary.



3.2 Level-0 Data Flow Diagram (Context Diagram)



3.3 Functional Requirements Specification

Req ID	Module	Requirement Description	Priority
FR-1.1	Auth	The system shall authenticate users via JWT and enforce Role-Based Access Control (RBAC).	High
FR-2.1	Farm Logic	Users shall create farms and subdivide them into fields specifying area (acres) and soil type.	High
FR-3.1	Rotation	The system shall suggest the top 3 optimal crops for a field based on crop family history and nitrogen rules.	High
FR-4.1	Operations	Creating a crop plan shall auto-generate field operations (sowing, irrigation, harvest) with worker mapping.	Medium
FR-5.1	Finances	The system shall track costs/harvests and compute Net Profit, Cost/Acre, and Yield/Acre.	High
FR-6.1	Equipment	A scheduled background job shall mark machinery as DUE/OVERDUE and trigger email notifications.	Medium

4. Database Design & Entity-Relationship Model

4.1 Normalized Entity-Relationship Diagram (3NF)

The relational database schema is normalized to 3rd Normal Form (3NF). The primary central entity is CROP_PLAN, connecting FIELD and CROP master data to FIELD_OPERATION, EXPENSE, and HARVEST tables.

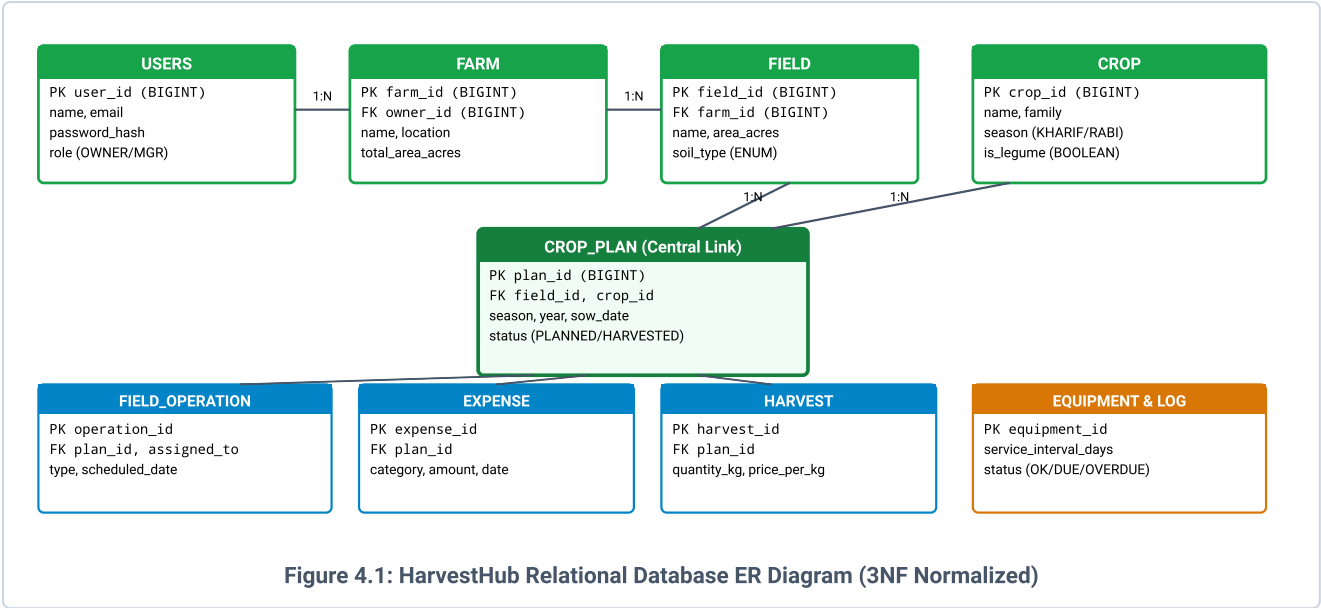


Figure 4.1: HarvestHub Relational Database ER Diagram (3NF Normalized)

4.2 Relational Data Dictionary

Table Name	Primary Key	Foreign Keys	Key Attributes & Constraints
users	user_id	None	email (UNIQUE, NOT NULL), password_hash, role (ENUM)
farm	farm_id	owner_id → users.user_id	name, location, total_area_acres (DECIMAL)
field	field_id	farm_id → farm.farm_id	area_acres, soil_type (ALLUVIAL/BLACK/CLAY/LOAMY)
crop_plan	plan_id	field_id, crop_id	UNIQUE(field_id, season, year), sow_date, status
equipment	equipment_id	farm_id → farm.farm_id	service_interval_days, last_service_date, status

5. Algorithmic Logic & REST API Contracts

5.1 Rule-Based Crop Rotation Algorithm

The crop recommendation service evaluates candidate crops for a target season based on agricultural logic:

Crop Scoring Formula

Score = Base(100) - SameCropPenalty(50) - SameFamilyPenalty(30) - RepeatPenalty(10 * count) + LegumeBonus(+25) + ProfitScore(0..20)

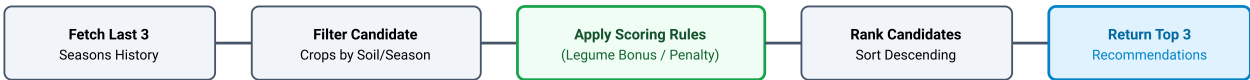


Figure 5.1: Algorithmic Recommendation Pipeline

5.2 REST Interface Specifications

HTTP Method	API Endpoint	Required Role	Functionality / Description
POST	/api/auth/login	Public	Authenticates credentials; returns signed JWT Bearer token.
GET	/api/farms/{id}/fields	Owner / Manager	Retrieves all fields and current soil profiles for a farm.
GET	/api/fields/{id}/rotation	Manager	Executes rotation engine; returns top 3 ranked crops with explanations.
POST	/api/plans	Manager	Creates a crop plan and triggers operational schedule auto-generation.
POST	/api/plans/{id}/expenses	Manager	Logs operational expense line item under specified category.
GET	/api/reports/profitability	Owner / Manager	Computes Net Revenue, Cost/Acre, and Yield metrics per season.

5.3 Sample API JSON Response Payload

```
{
  "fieldId": 102,
  "fieldName": "North Plot F-2",
  "targetSeason": "KHARIF_2026",
  "suggestions": [
    { "crop": "Moong (Green Gram)", "score": 138, "reason": "Legume after Wheat restores soil nitrogen; high farm historical profit." },
    { "crop": "Soybean", "score": 131, "reason": "Legume crop family; optimizes soil microbial diversity." },
    { "crop": "Maize", "score": 104, "reason": "Alternate crop family from previous Rabi cereal season." }
  ]
}
```

6. Non-Functional Requirements, Verification & Sign-Off

6.1 Quality Attributes & Non-Functional Requirements (NFRs)

Attribute	Target Performance Metric	Verification Strategy
Performance	API response times < 300ms for 95% of requests under concurrent load of 50 active users.	Apache JMeter load testing script execution.
Security	Zero plain-text passwords; stateless JWT auth; SQL injection prevention via Spring Data JPA parameterization.	OWASP ZAP dynamic security vulnerability scanning.
Reliability	99.5% operational uptime during active harvesting seasons; automated daily database backup.	Health endpoint monitoring via Spring Boot Actuator.
Portability	Fully responsive UI layout functioning on desktop (1080p) down to mobile viewports (360px).	Cross-browser testing on Chrome, Safari, and Firefox Mobile.

6.2 System Test Cases & Results

Test ID	Target Scenario	Input / Action	Expected Outcome	Status
TC-01	User Auth	Post invalid password to /api/auth/login	HTTP 401 Unauthorized with descriptive error.	PASS
TC-02	Area Validation	Add field area exceeding total farm capacity	System rejects request with validation error.	PASS
TC-03	Crop Rotation	Request suggestion for field with Wheat history	Legume crop (Moong) ranked #1 with nitrogen explanation.	PASS
TC-04	Maintenance Job	Set equipment last service date to -90 days	Status changes to OVERDUE and email alert fires.	PASS

6.3 Academic Review & Official Sign-Off

This Software Requirements Specification document for **HarvestHub** has been reviewed, evaluated, and approved by the academic supervisory panel of ABES Engineering College as fulfilling the baseline requirements for the Bachelor of Technology (B.Tech) CSE degree curriculum and HCL Industrial Training Program.

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